

Graduate Graph Theory 2026 Final

Assignment Solutions

- Do **five** of the following problems. Only one of the five can be a Bonus problem. (25 marks)
- Solutions, not just answers. Be clear and concise. I am marking these both on the ideas and on how you write them. If you know your handwriting is hard to read, please type your solutions.
- Due in by midnight on June 17. By hand or e-mail.

Problems

1. How many edges must you remove from the Petersen graph to make it planar?

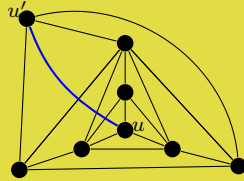
Solution

The Petersen graph has $n = 10$ vertices and $m = 15$ edges. If you remove one edge, Eulers formula says that an embedding has $f = 2 - n + m = 2 + 4 = 6$ faces. Each has at least 5 edges, so there must be $(6 + 5)/2 = 15 > 14$ edges. From the usual drawing of the Petersen graph, the first one in the notes, remove two non-incident edges in the inner C_5 , breaking it into a edge and a P_2 . The edge can be re-routed around the outside of the outer cycle.

2. The crossing number of a graph is the minimal number of edge crossings in a drawing of the graph in the plane. Show that the crossing number can be greater than the minimum number of edges you must remove from the graph to make it planar.

Solution

Take two K_3 s, one inside the other, and triangulate them. Triangulate them, then add a vertex u inside the inner K_3 and adjacent to all its vertices, and a vertex u' outside the outer one, adjacent to all its vertices. Add an edge from u to u' . This graph is 3-connected, so by Whitney, has a unique embedding—the embedding shown:



The edge uu' must cross both triangles, so the crossing number is at least 2, but removing this one edge makes the graph planar.

3. Find a graph G such that $\chi(G) = 2$ and $\chi_\ell(G) \geq 4$.

Solution

The graph $K_{10,10}$ has this property. Observe that there are 10 different three element subsets of the set $[5] = \{1, 2, 3, 4, 5\}$. Let each of these ten sets be the list on one vertex in each of the two partite sets. This is a set of lists of size 3, and it has no list colouring. Indeed, assume it does. We may assume that on one side, the vertex with list $\{1, 2, 3\}$ gets colour 1. The vertex with list $\{2, 3, 4\}$ gets some other colour $b \in \{2, 3, 4\}$, and there is another vertex with list $[5] \setminus \{1, b\}$ that gets a third colour. But then the vertex in the other partite set that has these three colours in its list cannot be coloured.

4. Show that an interval graph is a proper interval graph if and only if it does not contain an induced subgraph isomorphic to $K_{1,3}$.

Solution

Given any interval representation of a graph G , let the interval representing a vertex v be denoted $[L_v, R_v]$.

For the one direction, let G be an interval graph containing an induced $K_{1,3}$ with the vertex a adjacent to u, v and w . Assume an interval representation. We show that it cannot be proper. Indeed, as u, v, w are independent, we may assume that $L_u < R_u < L_v < R_v < L_w < R_w$. But then as a is adjacent to u and w we have $L_a < R_u$ and $L_w < R_a$. But then $[L_v, R_v] \subset [L_a, R_a]$, and so the representation is not proper. This was for any interval representation, so G is not a proper interval graph.

We now show that if G is an interval graph with no induced $K_{1,3}$ then it has a proper interval representation. Indeed, assume G is a interval graph with no induced $K_{1,3}$ and assume an interval representation that minimises the number of *bad pairs* (u, v) of vertices: vertices u and v for which $L_v \leq L_u < R_u \leq R_v$. If there are no bad pairs we are done, so we may assume, towards contradiction, that there is at least one.

For a bad pair (u, v) observe that if there is a non-neighbour a of u with $L_v < R_a < L_u$ then $a \sim v$; and if there is a non-neighbour b of u with $R_u < L_b < R_v$ then $b \sim v$. If both of these exist, then $\{u, v, a, b\}$ induces a $K_{1,3}$. So either $[L_v, L_u]$ contains no R_x or $[R_u, R_v]$ contains no L_x . In the former case we say that (u, v) has a *good left end of length* $|[L_v, L_u]|$ and in the latter case we say it has a *good right end of length* $|[R_u, R_v]|$.

Choose a bad pair (u, v) that minimises the size of its good end. We may assume wlog that this is a left end. As (u, v) has a good left end $[L_v, L_u]$ which contains no R_x . If (L_v, L_u) contains any L_x then x is in a bad pair with one of u or v with a shorter good left end, contradicting our choice of (u, v) . Thus there are no interval endpoints in (L_v, L_u) , and so we can replace the interval for u with $[L_v - \varepsilon, R_u]$ for small enough ε and get a new interval representation of G in which the interval for u is not contained in the interval for v . and no new bad pairs have been introduced. This contradicts the fact that our original representation minimised the number of bad pairs, and so there are no bad pairs, and so G is proper interval.

5. A vertex v in a chordal graph G is a *simplicial keystone* if there is a simplicial ordering of G in which v is the last vertex. What is that maximum d such that any large enough chordal graph has d simplicial keystones.

Solution

The maximum such d is 2. A simplicial keystone is necessarily simplicial. Arbitrarily long paths are chordal, and have only two simplicial vertices. Thus d can be at most 2. Let G be a chordal graph on at least 2 vertices. We showed that a Lex-BFS starting at any vertex will give a simplicial ordering, the final vertex of this is a simplicial keystone. As we can choose any vertex to be the initial vertex, including a simplicial keystone, to start a LEX-BFS from, and so excluding this vertex from being the final vertex, there must be at least one more simplicial keystone. So $d = 2$.

6. Show that if $p = o(\frac{\log n}{n})$ then a.a.s. $G_{n,p}$ has isolated vertices.

Solution

Sorry! This one uses second moment method and Chebyshev's inequality. We didn't get to this. It was not a fair question to ask, so I accepted proofs using any known results. Well done, any of you that figured it out.

Where X_v is the event that the vertex v is isolated, we have $E(X_v) = (1-p)^{n-1}$ as the $n-1$ edges that v is in must not be in $G_{n,p}$. By the linearity of expectation, the random variable $X = \sum_{v \in V(G)} X_v$ that counts number of isolated vertices has expected value

$$\begin{aligned} E(X) &= n(1-p)^{n-1} > ne^{-pn} > ne^{-\frac{\log(n)}{2}} \\ &= n(e^{\log n})^{-1/2} = n^{1-1/2} = \sqrt{n}. \end{aligned}$$

As X_v is Bernoulli, its variance is

$$\text{Var}(X_v) = (1-p)^{n-1}(1 - (1-p)^{n-1}) < (1-p)^{n-1} = E(X_v).$$

The X_v are not independent, but are close. The covariance of two is

$$\begin{aligned} \text{Cov}(X_u, X_v) &= E(X_u X_v) - E(X_u)E(X_v) \\ &= (1-p)^{2n-3} - (1-p)^{2n-2} = p(1-p)^{2n-2}. \end{aligned}$$

So the variance of X is

$$\begin{aligned} \text{Var}(X) &= \sum_u \text{Var}(X_u) + \sum_{u \neq v} \text{Cov}(X_u, X_v) \\ &\leq n(1-p)^{n-1} + n(n-1)p(1-p)^{2n-2} \\ &= E(X) + E(X)^2 \frac{n-1}{n} \cdot \frac{p}{1-p} \end{aligned}$$

By Chebyshev's inequality we then get

$$\begin{aligned} P(X=0) &\leq P(|X - E(X)| > E(X)) \leq \frac{\text{Var}(X)}{E(X)^2} \\ \text{(by Chebyshev)} &< \frac{1}{E(X)} + \frac{n-1}{n} \cdot \frac{p}{1-p} \rightarrow \frac{1}{E(x)} < \frac{1}{\sqrt{n}} \rightarrow 0 \end{aligned}$$

as $n \rightarrow \infty$ using in the end that $\frac{n-1}{n} \rightarrow 1$ and, as $p \rightarrow 0$, is small, that $\frac{p}{1-p} \rightarrow 0$. That is, as $n \rightarrow \infty$ the probability that there are no isolated vertices goes to 0, so a.a.s, there are isolated vertices.

Bonus

7. Make up, and solve, a problem that is non-trivial and appropriate to the notes. (If you find the problem somewhere, reference it.)

Solution

8. Find a mathematical mistake in the notes, and fix it. There are plenty, but if you can't find one, find a part of the notes that is unclear, and rewrite it to make it clear.

